

FORT FRANCES, ON, Canada

WMO: 719620

Lat: 48.6511N

Lon: 93.4389W

Elev: 342

StdP: 97.28

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.1	(c) -30.9	(d) -37.9	(e) 0.1	(f) -34.0	(g) -34.5	(h) 0.2	(i) -30.8	(j) 9.9	(k) -10.6	(l) 9.0	(m) -11.5	(n) 1.5	(o) 280	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	29.1	20.4	27.5	19.2	26.0	18.4	22.1	27.0	20.9	25.5	19.8	24.2	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.8	24.8	19.3	14.7	23.5	18.2	13.6	22.5	66.6	27.2	62.3	25.3	58.4	24.3	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	8.0	7.1	-38.8	31.8	3.4	0.9	-41.3	32.5	-43.3	33.0	-45.2	33.5	-47.7	34.1
			-38.9	24.0	3.4	1.2	-41.4	24.8	-43.4	25.5	-45.3	26.1	-47.8	27.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-15.4	-13.7	-4.5	3.1	10.4	15.9	18.7	17.4	13.0	5.8	-2.7	-12.1		
	DBStd	13.37	8.93	7.56	8.08	5.31	4.83	3.48	3.10	3.42	4.49	4.69	6.57	7.70		
	HDD10.0	3408	786	663	453	213	54	2	0	0	20	148	383	685		
	HDD18.3	5668	1044	897	708	457	249	87	34	59	167	389	632	943		
	CDD10.0	884	0	0	3	6	67	180	269	231	111	18	1	0		
	CDD18.3	103	0	0	0	0	4	14	45	31	8	1	0	0		
	CDH23.3	1133	0	0	2	1	78	172	464	327	82	8	0	0		
	CDH26.7	218	0	0	0	0	14	22	99	69	13	1	0	0		
(14) Wind	WSAvg	3.3	3.3	3.3	3.4	3.8	3.5	3.2	2.9	2.8	3.2	3.6	3.4	3.1		
(15) Precipitation	PrecAvg	743	29	23	32	52	75	115	105	91	89	61	35	33		
	PrecMax	988	87	45	78	127	187	222	269	195	194	137	99	52		
	PrecMin	581	7	5	5	2	9	13	34	31	18	9	4	12		
	PrecStd	94	18	10	18	31	46	54	57	36	46	39	19	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	5.9	19.4	20.9	28.8	29.0	31.1	30.7	28.5	23.9	16.0	4.8		
		MCWB	1.7	3.4	12.8	10.6	16.8	19.9	21.8	21.8	19.6	15.0	10.0	2.0		
	2%	DB	1.5	2.9	12.8	17.8	25.7	26.9	29.3	28.7	25.8	19.8	11.9	2.1		
		MCWB	0.3	0.6	7.8	9.3	15.1	18.2	21.2	20.9	18.4	12.9	7.6	0.8		
	5%	DB	0.1	0.6	9.0	15.1	22.9	25.4	27.5	26.8	23.5	16.6	8.8	0.6		
		MCWB	-0.9	-0.8	4.8	7.8	13.8	17.4	19.7	19.4	17.5	11.5	6.2	-0.2		
	10%	DB	-2.5	-1.8	5.7	12.7	20.3	23.7	25.9	25.0	21.3	13.7	6.3	-1.3		
		MCWB	-3.4	-3.3	2.6	6.7	13.0	16.8	19.0	18.5	16.6	9.9	4.2	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	3.2	15.4	13.1	19.0	22.0	24.1	23.1	21.8	15.8	11.2	2.6		
		MCDB	3.6	5.9	18.4	17.7	25.5	26.4	29.3	28.3	26.1	20.9	13.8	3.8		
	2%	WB	0.5	1.1	8.3	10.1	17.4	20.0	22.2	21.9	19.8	13.8	8.4	1.0		
		MCDB	1.5	2.4	12.6	15.8	22.0	24.6	26.9	27.4	23.4	19.2	11.4	1.9		
	5%	WB	-1.1	-0.8	5.0	8.5	15.7	18.8	21.1	20.7	18.5	12.1	6.3	-0.2		
		MCDB	0.1	0.5	8.2	14.4	21.0	23.3	25.8	25.3	22.6	14.9	8.9	0.7		
	10%	WB	-3.4	-3.2	2.8	7.0	14.0	17.7	20.0	19.6	17.1	10.2	4.0	-2.2		
		MCDB	-2.5	-2.0	5.5	12.3	19.3	22.2	24.4	23.7	20.6	13.4	6.0	-1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.9	12.4	11.5	12.8	11.8	12.5	12.8	11.7	9.5	8.1	9.1		
		MCDBR	9.8	13.0	13.8	16.8	18.3	15.0	14.7	14.8	14.8	15.1	11.9	8.0		
		MCWBR	8.6	10.6	9.2	9.9	9.2	8.0	7.9	8.0	8.9	9.2	8.2	7.0		
		MCDBR	9.4	10.7	12.7	14.6	13.4	12.2	12.3	12.3	12.6	12.0	11.3	7.9		
	5% WB	MCWBR	8.3	8.7	8.9	9.2	7.7	7.4	7.4	7.4	8.2	7.9	8.1	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.309	0.340	0.356	0.374	0.385	0.373	0.355	0.305	0.284	0.254		
	taud		2.363	2.322	2.361	2.402	2.414	2.385	2.377	2.442	2.462	2.585	2.503	2.402		
	Ebn at Noon		834	895	909	910	905	887	870	867	849	851	790	782		
	Edh at Noon		66	87	101	108	111	115	115	103	92	69	59	56		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.49	3.73	4.79	5.27	5.57	5.87	4.96	3.65	2.11	1.16	0.91		
	RadStd		0.17	0.25	0.34	0.42	0.35	0.44	0.34	0.35	0.31	0.29	0.14	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page